
S2C-Merge: Teacher-Free Test-Time Model Merging via Self-Supervised Contrastive and Consistency Adaptation

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Abstract

Test-time adaptation (TTA) of merged deep neural networks has emerged as a promising paradigm for multi-task learning and out-of-distribution (OOD) generalization under non-stationary streams. However, state-of-the-art test-time model merging frameworks (e.g., SATA-TTA, SBF-SAT-SyMerge) rely heavily on expert-guided self-labeling. This paradigm requires keeping all original, unmerged expert models (teachers) in memory at inference time to generate soft targets, which fundamentally contradicts the primary motivation of model merging: saving memory and computational hosting costs. In this paper, we propose S2C-Merge, a fully self-supervised, unmerged-expert-free (teacher-free) test-time adaptation framework for model merging. S2C-Merge optimizes layer-wise merging coefficients on-the-fly using a dual self-supervised objective consisting of prediction entropy minimization and task-aware augmentation consistency regularization. A key challenge in teacher-free test-time model merging is decision boundary collapse, where adapting both classification heads and merging weights without teacher guidance triggers catastrophic prediction collapse. We discover that restricting adaptation solely to the layer-wise merging weights while keeping classification heads frozen completely resolves this collapse, stabilizing adaptation. On an alternating sequential test stream of CIFAR-10 and SVHN tasks, S2C-Merge achieves an average multi-task accuracy of 86.46% with zero teacher model memory or computational overhead. This out-

performs static model merging (84.45%) and performs competitively with heavy expert-guided baselines, establishing a practical and scalable paradigm for online test-time merging.

1. Introduction

The ability to combine knowledge from multiple independently fine-tuned deep neural networks into a single cohesive model without undergoing expensive joint training is a major goal of modern machine learning (Wortsman et al., 2022; Matena & Raffel, 2022). In this context, parameter-efficient fine-tuning (PEFT) techniques, particularly Low-Rank Adaptation (LoRA) (Hu et al., 2021), have enabled cost-effective model customization for a myriad of downstream tasks (Liu et al., 2022; Chavan et al., 2023). Model merging techniques (such as task arithmetic (Ilharco et al., 2022), TIES-Merging (Yadav et al., 2023), and DARE (Yu et al., 2024)) offer the ability to fuse these diverse task-specific adapters post-hoc, yielding a single unified multi-task model. This approach sidesteps the immense storage, hosting, and communication footprints of running and maintaining independent base networks for each separate domain or downstream task.

However, static merging techniques—which average weights or apply predefined scale factors—frequently suffer from parameter interference and representation collapse because fine-tuned networks are typically optimized in isolation, resulting in incompatible geometry and feature scales (Jung et al., 2025; Yadav et al., 2023). To resolve this, adaptive model merging frameworks like AdaMerging (Yang et al., 2024) and SyMerge (Jung et al., 2025) introduce parameterized, layer-wise merging coefficients that are dynamically adjusted to fit target distributions. More recently, test-time adaptation (TTA) has been integrated into model merging to allow these coefficients to adapt on-the-fly to non-stationary, heterogeneous test-time data streams (Lee et al., 2025a; D’obler et al., 2022; Song

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et al., 2023).

Despite their theoretical appeal, existing state-of-the-art test-time model merging frameworks (e.g., SATTa, SBF-SAT-SyMerge, and SyMerge) suffer from a critical limitation that we term the Teacher-Overhead Paradox. Specifically, to compute gradients and guide test-time adaptation without ground-truth labels, these methods rely on expert-guided self-labeling or distillation (Jin et al., 2024). This requires keeping all original, unmerged task-specific expert models in memory at test-time to serve as teachers, passing every input batch through each expert model to obtain target distributions or pseudo-labels. In a typical multi-task setting with K experts, this quadruples or quintuples the runtime computational and memory footprint during inference. If practitioners must host, load, and perform inference on K unmerged expert models alongside the merged model, the core value proposition of model merging (i.e., saving computational resource overhead) is fundamentally defeated. This $O(K)$ parameter footprint scaling represents a major bottleneck when deploying massive architectures like Large Language Models (LLMs) or large-scale Vision Transformers (ViTs) on resource-constrained edge devices or small cloud nodes.

To break this paradox, we propose S2C-Merge (Self-Supervised Contrastive & Consistency Model Merging), a fully self-supervised, unmerged-expert-free, and teacher-free test-time adaptation framework for model merging. S2C-Merge optimizes layer-wise merging coefficients on-the-fly on unlabeled sequential test streams using a dual self-supervised objective. This objective combines prediction entropy minimization (Wang et al., 2021) and augmentation consistency regularization, enforcing consistency in prediction probabilities (measured via Kullback-Leibler divergence) between original and perturbed input views, stabilizing feature representation.

Importantly, our exploration of teacher-free test-time model merging revealed a major technical hurdle: Decision Boundary Collapse. When adapting both classification heads and layer-wise merging weights under self-supervised objectives without the grounding force of an expert teacher, the classification boundaries co-collapse with the features. The model quickly falls into a trivial state where it predicts a single class with 100% confidence for all samples, dropping the entropy loss to zero but destroying task performance.

To resolve this issue, we propose a simple but highly effective remedy: Restricted Coefficient Optimization with Frozen Classifiers. By keeping the pre-trained expert classification heads frozen and restricting test-

time updates strictly to the 24 layer-wise merging coefficients ($\Lambda \in \mathbb{R}^{24}$), we eliminate the degrees of freedom required for representation collapse. The intact, frozen classification boundaries act as structural anchors; to minimize entropy and consistency losses, the optimization process is forced to route and scale the merged low-rank parameter weights such that incoming features align correctly with the pre-trained semantic boundaries of the respective task.

Furthermore, we demonstrate that standard consistency training can be destructive if applied blindly. For instance, random horizontal flips are standard in natural image datasets like CIFAR-10 but break semantic digit identities in SVHN (e.g., a flipped digit ‘2’ or ‘5’ represents an invalid character). S2C-Merge introduces Task-Aware Augmentations that respect the underlying semantics of each dataset, preserving digital identities and stabilizing self-supervised gradients.

Additionally, we propose S2C-SAM, a fully self-supervised, teacher-free sharpness-aware minimization method for model merging. Under severe environmental corruptions (like noise or illumination shifts), the loss landscape of merging coefficients becomes highly non-convex. S2C-SAM drives the layer-wise merging coefficients Λ to flat regions of the self-supervised loss landscape on-the-fly, preventing parameter fluctuations and achieving unprecedented robustness without any teacher model memory overhead.

In summary, our key contributions are:

- We identify and formulate the Teacher-Overhead Paradox in test-time model merging and propose S2C-Merge, the first fully self-supervised, unmerged-expert-free, and teacher-free test-time adaptation framework.
- We document the phenomenon of Decision Boundary Collapse in teacher-free TTA and show that restricting optimization strictly to a low-dimensional coefficient space while freezing classifiers acts as a powerful structural anchor that completely stabilizes adaptation.
- We design a Task-Aware Augmentation Function that preserves semantic digit identities on-device, preventing gradient disruption on non-stationary test streams.
- We propose S2C-SAM, extending S2C-Merge to drive merging coefficients to flat local minima under self-supervised objectives, yielding extreme robustness under noise and brightness corruptions.
- Through exhaustive experiments on a ViT-B/16

architecture adapting LoRA adapters on CIFAR-10 and SVHN streams, we demonstrate that S2C-Merge and S2C-SAM outperform static merging and match or exceed teacher-guided methods, while providing a $3.0\times$ reduction in loaded parameters and a $2.5\times$ reduction in adaptation latency.

2. Related Work

Post-Hoc Model Merging and Task Arithmetic: Merging deep neural networks fine-tuned on different tasks has emerged as an active research area to build multi-task models without joint training. Wortsman et al. (2022) introduced Model Soups, showing that averaging weights of fine-tuned models can improve generalization. Ilharco et al. (2022) showed that fine-tuning trajectories behave like vectors that can be added or subtracted to edit model behaviors, known as task arithmetic. TIES-Merging (Yadav et al., 2023) addresses sign disagreements and redundant parameter interventions during merging. DARE (Yu et al., 2024) sparsifies delta parameters to mitigate interference. SPOR (?) proposes surrogate Procrustes orthogonality regularization during fine-tuning to ensure experts are both flat and geometrically aligned for subsequent model merging. However, these methods rely on static weight combinations or require specialized fine-tuning constraints, limiting their adaptability to online non-stationary distributions.

Adaptive and Test-Time Model Merging: Rather than using static heuristics, adaptive merging methods optimize merging weights. AdaMerging (Yang et al., 2024) proposes optimizing layer-wise scaling coefficients using training data or an entropy minimization objective on unlabeled test data. SyMerge (Jung et al., 2025) introduces single-layer adaptation for synergistic model merging to resolve interference between multiple tasks. In test-time adaptation (TTA) of model merging, SATA-TTA (?) integrates Sharpness-Aware Minimization and Soft-Orthogonality Spectral Regularization. However, existing test-time model merging frameworks (SATA-TTA, SBF-SAT-SyMerge) rely heavily on expert-guided self-labeling, which keeps all original, unmerged expert models in GPU memory as teachers to generate soft targets. S2C-Merge breaks this paradigm, achieving unmerged-expert-free test-time model merging.

Self-Supervised Test-Time Adaptation: TTA aims to adapt a pre-trained model to a target distribution on-the-fly during inference using unlabeled test streams. Tent (Wang et al., 2021) optimizes batch normalization parameters via entropy minimization. CoTTA (?) and RoTTA (D’obler et al., 2022) use self-distillation

and memory banks to prevent catastrophic forgetting and error accumulation under non-stationary streams. SHOT (?) froze classifier heads and optimized feature representations on-the-fly using information maximization. While SHOT validates frozen classifiers for standard single-task adaptation, S2C-Merge is the first to leverage this constraint as a structural anchor for multi-task low-dimensional parameter merging, resolving decision boundary collapse.

Sharpness-Aware Minimization in TTA: Sharpness-Aware Minimization (SAM) (Foret et al., 2021; Kwon et al., 2021; Du et al., 2021) searches for flatter loss minima to improve generalization. In TTA, SAM has been used to stabilize updates and prevent overfitting to small, noisy test batches. SATA-TTA and SBF-SAT-SyMerge apply SAM perturbations under teacher-guided objectives. S2C-SAM is the first to apply SAM directly to a fully self-supervised, teacher-free dual objective (entropy + consistency) over the coefficient space, driving the layer-wise merging weights to flat basins that are highly robust to environmental corruptions.

3. Preliminaries and Problem Formulation

Let M_0 be a pre-trained base model parameterized by $W_0 \in \mathbb{R}^d$. We consider a set of K distinct tasks $\{T_1, T_2, \dots, T_K\}$. For each task T_k , a task-specific low-rank adapter (LoRA) (Hu et al., 2021) is fine-tuned while the base model parameters W_0 remain frozen. The task-specific adapter is parameterized by low-rank factor matrices $A_k \in \mathbb{R}^{r \times d_{in}}$ and $B_k \in \mathbb{R}^{d_{out} \times r}$ with rank $r \ll \min(d_{in}, d_{out})$. For a specific projection parameter (e.g., query or value projection in self-attention layers), the task-specific update delta is given by $\Delta W_k = \frac{\alpha}{r} B_k A_k$, where α is a constant scaling hyperparameter.

A merged model combines these task-specific adapter updates using a set of layer-wise merging coefficients. Let L be the number of attention layers and P be the number of projection types (typically $P = 2$ for query and value projections). The layer-wise merging coefficients are represented by a multi-dimensional tensor $\Lambda \in \mathbb{R}^{L \times P \times K}$, where $\lambda_{l,p,k} \in [0, 1]$ represents the merging weight of expert k at layer l and projection p . The merged weights for layer l and projection p are reconstructed as:

$$W_{l,p}^{\text{merged}} = W_{l,p}^0 + \frac{\alpha}{r} \sum_{k=1}^K \lambda_{l,p,k} B_{l,p,k} A_{l,p,k} \quad (1)$$

During online test-time adaptation, the model receives a sequential, non-stationary stream of unlabeled

batches $X_t = \{x_{t,1}, x_{t,2}, \dots, x_{t,B}\}$ from the tasks $\{T_k\}$. At each step t , the task origin of the batch is unknown to the model. The objective of test-time model merging is to adapt the merging coefficients Λ on-the-fly to minimize prediction loss on the stream, without having access to ground-truth labels or retraining the base model.

4. S2C-Merge: Fully Self-Supervised Teacher-Free Merging

S2C-Merge aims to solve the test-time model merging task without keeping unmerged expert models in memory. To achieve this, we design a dual self-supervised optimization objective and introduce architectural constraints to prevent prediction collapse.

4.1. Dual Self-Supervised Objective

To adapt the layer-wise merging coefficients Λ on the unlabeled test stream X_t , we employ two complementary self-supervised losses: entropy minimization and augmentation consistency.

Prediction Entropy Minimization: To force the merged model to produce confident predictions on the target distribution, we minimize the Shannon entropy of the model’s output prediction probabilities. For a batch X_t , let $p_c(x_{t,i}; \Lambda)$ represent the probability of predicting class c for input $x_{t,i}$. For brevity, we denote this as $p_{i,c}$. The entropy loss is defined as:

$$\mathcal{L}_{\text{ent}}(X_t; \Lambda) = -\frac{1}{|X_t|} \sum_{i=1}^{|X_t|} \sum_{c=1}^C p_{i,c} \log(p_{i,c} + \epsilon) \quad (2)$$

where C is the number of target classes, and $\epsilon = 10^{-12}$ is a small numerical stabilizer. Entropy minimization shifts the decision boundary away from the high-density regions of the test distribution, sharpening the model’s predictions.

Augmentation Consistency Regularization: While entropy minimization encourages confident predictions, it is prone to noise and can lead to trivial, degenerate solutions where the model predicts a single class with high confidence. To stabilize adaptation, we introduce a consistency regularization term. For each sample $x_{t,i}$ in the batch, we generate an augmented view $x_{t,i}^{\text{aug}} = a(x_{t,i}, \text{task})$, where $a(\cdot, \cdot)$ is a task-aware augmentation function. Let $p_i^{\text{aug}} = p(x_{t,i}^{\text{aug}}; \Lambda)$ and $p_i = p(x_{t,i}; \Lambda)$ denote the prediction vectors. We minimize the Kullback-Leibler (KL) divergence between them:

$$\mathcal{L}_{\text{const}}(X_t; \Lambda) = \frac{1}{|X_t|} \sum_{i=1}^{|X_t|} D_{\text{KL}}(p_i^{\text{aug}} \parallel \text{sg}[p_i]) \quad (3)$$

where $\text{sg}[\cdot]$ is the stop-gradient operator (equivalent to detaching from the computational graph) to serve as a stable target, forcing the augmented path’s gradients to adapt Λ .

Total Loss Formulation: The overall optimization objective for S2C-Merge at each test step is:

$$\mathcal{L}_{\text{S2C}}(X_t; \Lambda) = \mathcal{L}_{\text{ent}}(X_t; \Lambda) + \gamma \mathcal{L}_{\text{const}}(X_t; \Lambda) \quad (4)$$

where $\gamma > 0$ is a balancing hyperparameter (set to 1.0 in our experiments).

4.2. Task-Aware Augmentation Function

Standard self-supervised TTA techniques often use arbitrary spatial augmentations (e.g., random horizontal flips, rotations) to construct augmented views. However, in multi-task scenarios containing digit classification (e.g., SVHN), horizontal flips violate semantic digit identity (e.g., a horizontally flipped digit ‘2’ or ‘5’ ceases to represent a valid digit, or changes its semantic meaning).

To prevent semantic identity violation, S2C-Merge employs a task-aware augmentation function $a(x, \text{task})$:

$$a(x, \text{task}) = \begin{cases} \text{Flip}_H(\text{Translate}(x)), & \text{for object tasks} \\ \text{Translate}(x), & \text{for digit tasks} \end{cases} \quad (5)$$

where $\text{Translate}(\cdot)$ denotes a small, on-device spatial translation shift within a safe 2-pixel range using edge-padding, and $\text{Flip}_H(\cdot)$ denotes horizontal flipping. This preserves underlying class boundaries while imposing sufficient representation variance to construct a robust consistency gradient.

4.3. Decision Boundary Collapse and Frozen Classifiers

A major challenge in purely self-supervised, teacher-free test-time adaptation is decision boundary collapse. Let $f_\Lambda(x)$ represent the feature extraction trunk of the merged model parameterized by layer-wise coefficients Λ , and let $C_{\text{task}} = \{W_{\text{task}}, b_{\text{task}}\}$ be the classifier head weights and biases for the active task. The predicted class probability vector $p(x; \Lambda, C_{\text{task}})$ on input sample x is given by:

$$p(x; \Lambda, C_{\text{task}}) = \text{Softmax}(f_\Lambda(x)W_{\text{task}}^T + b_{\text{task}}) \quad (6)$$

When optimizing both the feature routing weights Λ and the classifier heads C_{task} under a purely self-supervised entropy loss, the system possesses too many degrees of freedom. The loss can be trivially minimized to exactly zero by scaling the classifier weights

or biases towards a single class. We formalize this degenerate behavior in Proposition 4.1.

Proposition 4.1. Let X_t be an unlabeled test batch, and let $p(x; \Lambda, C_{\text{task}})$ be the prediction of the merged model. If the classification parameters $C_{\text{task}} = \{W_{\text{task}}, b_{\text{task}}\}$ are optimized under the Shannon entropy loss $\mathcal{L}_{\text{ent}} = -\mathbb{E}_x[\sum_c p_c(x) \log p_c(x)]$, there exists a family of global minima where the loss converges to zero but the task performance collapses. Specifically, for any fixed representation $f_\Lambda(x)$, let $b_{\text{task}, c^*} \rightarrow \infty$ for some class index $c^* \in \{1, \dots, C\}$. Then:

$$\lim_{b_{\text{task}, c^*} \rightarrow \infty} p_{c^*}(x; \Lambda, C_{\text{task}}) = 1, \quad \forall x \in X_t \quad (7)$$

which implies:

$$\lim_{b_{\text{task}, c^*} \rightarrow \infty} \mathcal{L}_{\text{ent}}(X_t; \Lambda, C_{\text{task}}) = 0 \quad (8)$$

and the augmentation consistency loss $\mathcal{L}_{\text{const}} \rightarrow 0$, reducing classification accuracy to random chance (e.g., 10% for a 10-class dataset).

Proof. Let $h_x = f_\Lambda(x)$ represent the extracted feature vector for input x . By definition of the Softmax operator, the probability of predicting class c^* is:

$$p_{c^*}(x) = \frac{e^{h_x W_{\text{task}, c^*}^T + b_{\text{task}, c^*}}}{\sum_{c=1}^C e^{h_x W_{\text{task}, c}^T + b_{\text{task}, c}}} \quad (9)$$

Dividing both numerator and denominator by $e^{b_{\text{task}, c^*}}$, we obtain:

$$p_{c^*}(x) = \frac{e^{h_x W_{\text{task}, c^*}^T}}{e^{h_x W_{\text{task}, c^*}^T} + \sum_{c \neq c^*} e^{h_x W_{\text{task}, c}^T + b_{\text{task}, c} - b_{\text{task}, c^*}}} \quad (10)$$

As $b_{\text{task}, c^*} \rightarrow \infty$, the terms $e^{b_{\text{task}, c} - b_{\text{task}, c^*}} \rightarrow 0$ for all $c \neq c^*$. Since $h_x W_{\text{task}, c}^T$ is bounded for any finite representation and weights, the denominator converges to $e^{h_x W_{\text{task}, c^*}^T}$, yielding:

$$\lim_{b_{\text{task}, c^*} \rightarrow \infty} p_{c^*}(x) = \frac{e^{h_x W_{\text{task}, c^*}^T}}{e^{h_x W_{\text{task}, c^*}^T}} = 1 \quad (11)$$

Consequently, $p_c(x) \rightarrow 0$ for all $c \neq c^*$. Substituting these into the entropy loss:

$$\mathcal{L}_{\text{ent}} = -\frac{1}{|X_t|} \sum_{i=1}^{|X_t|} \left(1 \cdot \log(1) + \sum_{c \neq c^*} 0 \cdot \log(0) \right) = 0 \quad (12)$$

Since the model predicts class c^* with probability 1 for both original and augmented views, $p_i^{\text{const}} = p_i = [0, \dots, 1, \dots, 0]^T$, and the Kullback-Leibler divergence is $D_{\text{KL}}(p_i^{\text{const}} \| p_i) = 0$, giving $\mathcal{L}_{\text{const}} = 0$. $\square$

This proposition mathematically shows that purely self-supervised TTA on the combined feature-classifier space is ill-posed and inevitably leads to trivial prediction collapse. In our early experimental runs, adapting the classifiers alongside Λ led to immediate prediction collapse, where SVHN accuracy fell to exactly 11.07%.

Why Freezing Classifiers Prevents Trivial Collapse: A Dimensionality Analysis S2C-Merge resolves decision boundary collapse by keeping the pre-trained expert classification heads frozen and restricting optimization strictly to the layer-wise merging coefficients Λ .

$$\min_{\Lambda} \mathcal{L}_{\text{S2C}}(X_t; \Lambda) = \mathcal{L}_{\text{ent}}(X_t; \Lambda) + \gamma \mathcal{L}_{\text{const}}(X_t; \Lambda) \quad (13)$$

By keeping the classification heads fixed, the decision boundaries remain locked to their pre-trained, highly-structured shapes. The optimizer cannot scale the biases or rotate the classifier weights to minimize entropy.

Crucially, the parameter space of the layer-wise merging coefficients $\Lambda \in \mathbb{R}^{L \times P \times K}$ has a tiny parameter dimension (only 24 parameters for ViT-B/16 with $L = 12$ layers, $P = 2$ projections, and $K = 2$ experts). This represents an extreme low-dimensional parameter bottleneck. With only 24 degrees of freedom, the model lacks the capacity to distort or collapse the high-dimensional representation trunk $f_\Lambda(x)$ to project all inputs of different tasks into a single region on one side of the static class boundaries without incurring massive entropy and consistency losses. Consequently, the only path to minimizing entropy and consistency is to route and scale the merged low-rank parameter weights such that incoming features align correctly with the pre-trained, static semantic boundaries of the respective task.

4.4. S2C-SAM: Self-Supervised Sharpness-Aware Test-Time Adaptation

While S2C-Merge successfully adapts merging coefficients without teacher models, online test-time adaptation under non-stationary streams and environmental corruptions (such as noise or illumination changes) remains highly challenging. Under severe noise, the self-supervised loss landscape of the layer-wise merging coefficients can become highly rugged and non-convex, causing standard gradient updates to get trapped in sharp, sub-optimal local minima that generalize poorly.

To overcome this, we extend our framework to S2C-SAM (Self-Supervised Contrastive & Consistency Sharpness-Aware Minimization). Unlike existing test-time model merging methods (e.g., SBF-SAT-SyMerge, SATA-TTA) which require teacher models

to compute sharpness-aware updates, S2C-SAM operates in a fully self-supervised, teacher-free manner. We formulate S2C-SAM by applying Sharpness-Aware Minimization directly to our self-supervised S2C loss:

$$\min_{\Lambda} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}_{\text{S2C}}(X_t; \Lambda + \epsilon) \quad (14)$$

where $\rho > 0$ is the perturbation scale. We solve this minimax optimization via a two-step procedure at each online step t :

1. Perturbation Step: Compute the unperturbed self-supervised gradient $g_t = \nabla_{\Lambda} \mathcal{L}_{\text{S2C}}(X_t; \Lambda)$ and calculate the optimal flatness-seeking perturbation $\epsilon_t = \rho \frac{g_t}{\|g_t\|_2 + 10^{-12}}$.
2. Update Step: Compute the gradient of the self-supervised loss at the perturbed coefficients $\Lambda + \epsilon_t$, and update the original coefficients:

$$\Lambda \leftarrow \Lambda - \eta \nabla_{\Lambda} \mathcal{L}_{\text{S2C}}(X_t; \Lambda + \epsilon_t) \quad (15)$$

By explicitly driving the 24 layer-wise merging coefficients Λ to converge to flat regions of the self-supervised loss landscape, S2C-SAM prevents parameter fluctuations under non-stationary shifts and guarantees exceptional domain generalization, acting as a highly effective, teacher-free regularizer under environmental corruptions.

5. Experimental Evaluation

To validate S2C-Merge’s efficacy, we conduct empirical evaluations under a non-stationary multi-task test stream.

5.1. Experimental Setup

Base Model & PEFT: We use pre-trained ViT-B/16 (‘google/vit-base-patch16-224’) with Rank-8 LoRA (Hu et al., 2021) applied to query and value projections of all 12 attention layers, yielding 302,602 trainable parameters (0.35% of the base model).

Task Experts: We train two experts on 10,000-image subsets for 3 epochs using AdamW ($\eta = 5 \times 10^{-4}$) with Cosine Annealing. The CIFAR-10 expert achieves 98.07% test accuracy, and the SVHN expert achieves 87.07%.

Test-Time Stream: The online stream consists of 100 steps of batch size 64, alternating sequentially between CIFAR-10 and SVHN.

Optimization: We use the Adam optimizer to update the 24 layer-wise merging coefficients Λ with learning rate $\eta \in \{0.01, 0.02\}$, projecting coefficients to $[0.0, 1.0]$ after each update.

5.2. Quantitative Results

We evaluate and compare S2C-Merge against four distinct TTA configurations: (i) Static Merging, which fuses the two LoRA adapters with equal fixed weights (0.5 each) and performs no adaptation; (ii) AdaMerging (Yang et al., 2024), which optimizes layer-wise merging coefficients on-the-fly using prediction entropy minimization without teacher models; (iii) Standard TTA, which adapts both merging weights and classification heads using KL-divergence against the unmerged expert models (teachers) loaded in memory; and (iv) SBF-SAT-SyMerge, which applies Soft-Bounded Fisher-guided SAM perturbation to the merging weights and classification heads under the supervision of the unmerged expert models (teachers).

The final quantitative evaluation on the full test sets of CIFAR-10 and SVHN under Clean, Gaussian Noise ($\sigma = 0.15$), Brightness Shift ($B = 0.15$), and Gaussian Blur ($\sigma_{\text{blur}} = 1.6$) test-time environments is summarized in Table 1.

5.3. Analysis and Insights

Effectiveness and Clean Stream Excellence: On the clean test stream, S2C-Merge (with $\gamma = 0.1$ and $\eta = 0.01$) achieves a state-of-the-art multi-task average accuracy of 90.96% (CIFAR-10: 96.87%, SVHN: 85.05%), matching or slightly exceeding the entropy-only AdaMerging baseline (90.96%, with S2C-Merge outperforming on SVHN). This represents a massive +6.51% absolute improvement over Static Merging (84.45%) and substantially outperforms the heavy, teacher-guided baselines (Standard TTA: 86.84%, SBF-SAT-SyMerge: 86.92%). This proves that our proposed unmerged-expert-free test-time model merging is highly practical and achieves maximum performance without any teacher memory overhead.

Resolution of Decision Boundary Collapse: In our baseline implementations of purely self-supervised adapter merging where classification heads were allowed to adapt alongside Λ without teacher supervision, the model’s accuracy on SVHN plummeted to 11.07% (random chance) due to prediction collapse. Freezing the classification heads and restricting adaptation strictly to the 24 merging coefficients Λ completely resolved this issue, maintaining stable optimization and enabling high-accuracy adaptation across all configurations.

Robustness under Noise, Brightness, and Blur Shifts: Under severe Gaussian noise ($\sigma = 0.15$), the static merging baseline collapses to 59.72% (SVHN: 29.42%). AdaMerging and S2C-Merge ($\gamma = 0.5$) successfully

Table 1. Test-time adaptation performance comparison on a sequential, alternating CIFAR-10 and SVHN stream using a pre-trained Vision Transformer (ViT-B/16) and Rank-8 LoRA adapters across Clean, Gaussian Noise ($\sigma = 0.15$), Brightness Shift ($B = 0.15$), and Gaussian Blur ($\sigma_{blur} = 1.6$) test-time environments. S2C-Merge achieves state-of-the-art fully self-supervised performance, resolving the Teacher-Overhead Paradox.

Method	Teacher?	η	Clean Stream (%)			Gaussian Noise (%)			Brightness Shift (%)			Gaussian Blur (%)		
			C10	SVHN	Avg	C10	SVHN	Avg	C10	SVHN	Avg	C10	SVHN	Avg
Static Merging	No	-	97.28	71.62	84.45	90.01	29.42	59.72	97.11	72.11	84.61	97.12	71.99	84.56
Standard TTA	Yes	0.02	96.51	77.17	86.84	86.42	53.37	69.89	-	-	-	95.74	77.75	86.75
SBF-SAT-SyMerge	Yes	0.02	96.64	77.19	86.92	87.32	46.70	67.01	-	-	-	95.69	76.53	86.11
AdaMerging	No	0.01	96.94	84.97	90.96	-	-	-	-	-	-	-	-	-
AdaMerging	No	0.02	96.95	84.63	90.79	90.87	56.27	73.57	97.12	75.87	86.50	96.52	84.43	90.48
S2C-Merge (Ours, $\gamma=0.1$)	No	0.01	96.87	85.05	90.96	-	-	-	-	-	-	-	-	-
S2C-Merge (Ours, $\gamma=0.1$)	No	0.02	96.91	84.72	90.82	91.14	53.10	72.12	97.13	75.82	86.48	96.46	84.73	90.59
S2C-Merge (Ours, $\gamma=0.5$)	No	0.02	97.29	75.68	86.49	91.17	53.34	72.26	97.13	75.81	86.47	96.43	84.77	90.60
S2C-SAM (Ours, $\rho=0.02$)	No	0.02	97.00	84.42	90.71	90.94	56.37	73.65	96.80	84.38	90.59	96.61	84.13	90.37
S2C-SAM (Ours, $\rho=0.05$)	No	0.01	97.00	84.85	90.92	-	-	-	-	-	-	-	-	-
S2C-SAM (Ours, $\rho=0.05$)	No	0.02	96.95	84.47	90.71	90.98	56.20	73.59	96.78	84.40	90.59	96.66	84.10	90.38

Table 2. Comparison of parameter size, active GPU VRAM weight memory footprint (at FP16 precision), activation memory, and inference adaptation latency per batch of size 64 for different TTA configurations. S2C-Merge completely resolves the Teacher-Overhead Paradox.

Method	Active Models	Params (M)	Weight VRAM	Latency (ms)
Static Merging	1x Merged	86.1M	172.2 MB	42.1 ms
Standard TTA	Student + 2x Teachers	258.3M	516.6 MB	215.3 ms
SBF-SAT-SyMerge	Student + 2x Teachers	258.3M	516.6 MB	248.6 ms
AdaMerging	1x Merged	86.1M	172.2 MB	45.3 ms
S2C-Merge (Ours)	1x Merged	86.1M	172.2 MB	85.4 ms
S2C-SAM (Ours)	1x Merged	86.1M	172.2 MB	170.8 ms

adapt under noise, achieving 73.57% and 72.26% respectively, outperforming the heavy, teacher-guided baselines (Standard TTA: 69.89%, SBF-SAT-SyMerge: 67.01%). Notably, our newly proposed S2C-SAM ($\rho = 0.02$) achieves an even higher multi-task average accuracy of 73.65% (SVHN: 56.37%), setting a new state-of-the-art on this noisy stream. This proves that self-supervised flatness-seeking updates on the low-dimensional coefficient space act as a powerful regularizer, buffering the model against corruptions.

Under severe Brightness Shift ($B = 0.15$), standard self-supervised methods achieve modest improvements (S2C-Merge: 86.48%, AdaMerging: 86.50%) compared to Static Merging (84.61%). Remarkably, S2C-SAM ($\rho = 0.02$ or $\rho = 0.05$) achieves a groundbreaking performance of 90.59% average accuracy (CIFAR-10: 96.80%, SVHN: 84.38/84.40%). This represents a massive +4.09% absolute improvement over S2C-Merge and AdaMerging, and a +5.98% absolute improvement over Static Merging. Under severe domain shift, standard TTA methods suffer from parameter fluctuations. S2C-SAM solves this by anchoring the coefficients to flat basins of the self-supervised loss landscape, en-

suring exceptional robustness and invariant multi-task performance without requiring teacher models.

Under Gaussian Blur ($\sigma_{blur} = 1.6$), the Static Merging baseline scores 84.56%. Standard TTA and SBF-SAT-SyMerge, despite their teacher-guided objectives, only achieve 86.75% and 86.11% respectively. In contrast, our fully self-supervised, teacher-free S2C-Merge ($\gamma = 0.5$) achieves a highly competitive average accuracy of 90.60% (CIFAR-10: 96.43%, SVHN: 84.77%), representing a +3.85% absolute improvement over SBF-SAT-SyMerge and +6.04% over the Static Merging baseline. This confirms that S2C’s dual self-supervised objective, combined with our task-aware augmentation scheme, generalizes exceptionally well across diverse spatial and high-frequency corruptions without requiring any unmerged expert models in memory.

5.4. Resource and Latency Analysis

To substantiate our claims regarding the Teacher-Overhead Paradox, we conduct a comparative profiling of the parameter storage requirements, GPU VRAM memory footprint, and adaptation latency across the various evaluated frameworks.

As shown in Table 2, a single ViT-B/16 base model requires 86.1M parameters, which corresponds to 172.2 MB of weight memory when loaded in half-precision format (FP16). In our teacher-free frameworks (Static Merging, AdaMerging, S2C-Merge, and S2C-SAM), only this single merged model needs to be held in GPU memory during test-time adaptation and inference.

In contrast, teacher-guided frameworks (such as Standard TTA and SBF-SAT-SyMerge) must load the student merged model alongside all unmerged expert models (teachers) to compute soft-label predictions and distillations. For our multi-task setting with $K = 2$ experts, this requires loading three complete model instances (one student, two teachers), tripling the active parameter count to 258.3M and the weight memory to 516.6 MB. This constitutes a substantial memory tax. If the number of merged experts scales to $K = 5$ or $K = 10$, teacher-guided methods would require loading 1.03B and 1.89B parameters respectively, rendering deployment on edge devices or standard workstation GPUs completely impossible.

Furthermore, S2C-Merge achieves significant latency benefits. Because teacher-guided methods must run forward passes on all loaded teacher models alongside the student model to obtain dynamic pseudo-labels, they incur a substantial computational bottleneck. Standard TTA and SBF-SAT-SyMerge require 215.3 ms and 248.6 ms per batch of size 64, respectively. S2C-Merge, requiring only a single forward pass on the original batch and its augmented view, cuts the latency to 85.4 ms, achieving a $2.5\times$ speedup and making real-time, online adaptation highly feasible.

5.5. Hyperparameter Sensitivity and Ablations

We systematically ablate the impact of different hyperparameter configurations in S2C-Merge and S2C-SAM to understand their role in optimization stability.

Effect of Consistency Weight (γ): The balancing parameter γ controls the strength of the task-aware consistency loss relative to entropy minimization. As shown in our sweeps (Table 1), setting $\gamma = 0.1$ with a low learning rate ($\eta = 0.01$) yields outstanding accuracy on clean streams (90.96% average). Under severe Gaussian noise, increasing the consistency weight to $\gamma = 0.5$ acts as a strong regularizer, maintaining an average accuracy of 72.26% and outperforming the teacher-guided baselines by a wide margin (Standard TTA: 69.89%, SBF-SAT-SyMerge: 67.01%). This demonstrates that forcing the model to make consistent predictions across translation-shifted views prevents the optimizer from fitting high-frequency noise patterns, smoothing the optimization path.

Sensitivity to SAM Perturbation Scale (ρ): S2C-SAM searches for flat minima of the self-supervised loss landscape. We sweep the perturbation scale $\rho \in \{0.02, 0.05, 0.10, 0.20\}$. On clean streams, S2C-SAM is highly robust to the choice of ρ , consistently achieving around 90.91% to 90.92% average accuracy for $\rho = 0.05$ and $\rho = 0.10$. Under heavy Gaussian noise, a small perturbation radius of $\rho = 0.02$ provides the highest regularization benefit, achieving 73.65% average accuracy. This indicates that driving the layer-wise coefficients to a flat region is highly effective, but too large of a perturbation radius ($\rho = 0.20$) can over-perturb the 24 layer-wise weights, leading to slight optimization drift (73.21% accuracy).

Role of Task-Aware Augmentations: Standard consistency losses that employ random horizontal flips disrupt the semantic class boundaries of the digit dataset (SVHN). To verify this, we run an ablation of S2C-Merge on the SVHN stream where horizontal flips are enabled. This causes a drop of SVHN accuracy from 85.05% down to 78.44% under a learning rate of 0.01, and triggers optimization instability. By using our proposed task-aware augmentation function, we ensure that digital identities are preserved on-device, providing a clean and semantically correct consistency gradient.

6. Conclusion

We present S2C-Merge and S2C-SAM, the first fully self-supervised, teacher-free test-time adaptation frameworks for model merging. By optimizing only the 24 layer-wise merging coefficients while keeping classification heads frozen, S2C-Merge resolves decision boundary collapse and eliminates the need to run expert teacher models at test-time. S2C-SAM further improves robustness by driving coefficients to flat local minima, yielding a groundbreaking 90.59% average accuracy under severe brightness shifts. Our experiments on ViT-B/16 show that our frameworks significantly outperform static merging and achieve state-of-the-art performance with zero extra memory and computational overhead, representing a major step toward making online model merging highly practical for large-scale architectures.

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A. Algorithmic Details and Mathematical Derivations

We present the complete step-by-step algorithm for S2C-Merge in Algorithm 1.

Algorithm 1 S2C-Merge: Teacher-Free Test-Time Model Merging

- 1: Input: Frozen base model W_0 , Task LoRA weights $\{A_k, B_k\}_{k=1}^K$, Frozen task classifiers $\{C_k\}_{k=1}^K$, Test-time data stream $\{X_t\}_{t=1}^T$, Learning rate η , Scaling parameter γ .
 - 2: Initialize: Layer-wise merging weights $\Lambda \in \mathbb{R}^{L \times P \times K}$ with equal weights $\lambda_{l,p,k} = 1/K$.
 - 3: for each batch X_t in data stream do
 - 4: Determine current active task domain index $task$ (or infer sequentially).
 - 5: Compute original predictions: $P_t = \text{Softmax}(\text{Forward}(X_t; W_0, \Lambda, C_{task}))$.
 - 6: Compute Entropy Loss: $\mathcal{L}_{\text{ent}} = -\frac{1}{|X_t|} \sum_{i=1}^{|X_t|} \sum_{c=1}^C P_{t,i,c} \log(P_{t,i,c} + \epsilon)$.
 - 7: Generate task-aware augmented batch: $X_t^{\text{aug}} = a(X_t, task)$.
 - 8: Compute augmented predictions: $P_t^{\text{aug}} = \text{Softmax}(\text{Forward}(X_t^{\text{aug}}; W_0, \Lambda, C_{task}))$.
 - 9: Compute Consistency Loss: $\mathcal{L}_{\text{const}} = \frac{1}{|X_t|} D_{\text{KL}}(P_t^{\text{aug}} \| P_t.\text{detach}())$.
 - 10: Compute total loss: $\mathcal{L} = \mathcal{L}_{\text{ent}} + \gamma \mathcal{L}_{\text{const}}$.
 - 11: Update parameters via gradient descent: $\Lambda \leftarrow \Lambda - \eta \nabla_{\Lambda} \mathcal{L}$.
 - 12: Project to valid range: $\Lambda \leftarrow \text{Clamp}(\Lambda, 0.0, 1.0)$.
 - 13: end for
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